

Experimental Results Outline

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1 Experimental Results

This section presents a comprehensive experimental evaluation of the proposed few-qubit QPP-QNN framework. We first describe the datasets, data splits, evaluation metrics, baseline methods, and training settings used to ensure a consistent comparison. We then evaluate the model on the clean synthetic benchmark and compare it with representative classical approaches and

other quantum model families. Its robustness is further examined under multiple image corruptions and noise intensities, followed by ablation studies of the main circuit-design choices. Finally, cross-domain transfer experiments and executions on quantum hardware are reported to assess the generality and practical feasibility of the proposed framework.

1.1 Experimental Setup

All methods were evaluated on a common image-level partition of the synthetic corner benchmark. The experiments formulate local keypoint detection as binary classification, where patches associated with L-corners, T-junctions, and X-junctions constitute the positive class and patches sampled away from any annotated keypoint constitute the negative class. The following protocol specifies the dataset construction, quantum input representation, reference methods, training procedure, and evaluation criteria used in the main experiments.

1.1.1 Datasets and Data Splits

The clean benchmark contains 300 grayscale images of size 64×64 , with 100 images generated for each of the L-corner, T-junction, and X-junction scene types. The rendered line width ranges from one to three pixels. To introduce moderate appearance variation without corrupting the underlying geometry, the image contrast is sampled from $[0.9, 1.1]$ and the brightness offset from $[-0.02, 0.02]$. No additive noise or blur is applied to the clean benchmark. Although the generator retains the original junction type, all three types are merged into a single keypoint class in the present binary-detection task.

Each image provides 25 candidate centers from which 9×9 patches are extracted. Five candidates are sampled by jittering the annotated keypoint and are assigned a positive label, whereas the remaining 20 candidates are sampled at least five pixels away from every annotated keypoint and are assigned a negative label. The resulting dataset therefore contains 7,500 patches, with 1,500 positive and 6,000 negative examples. Images, rather than individual patches, are partitioned in order to prevent patches from the same scene from appearing in different subsets. The split is stratified by scene type and uses 180 images for training, 60 for validation, and 60 for testing, corresponding to 4,500/1,500/1,500 patches and 900/300/300 positive samples, respectively. Dataset generation and splitting use random seed 37. All corrupted test sets in the robustness experiments are derived from the held-out test images while preserving their candidate centers and labels.

1.1.2 Model and Quantum Encoding

For each patch, image gradients are used to form the local structure tensor

$$\mathbf{M} = \begin{bmatrix} \sum I_x^2 & \sum I_x I_y \\ \sum I_x I_y & \sum I_y^2 \end{bmatrix}, \quad (1)$$

whose ordered eigenvalues are denoted by $\lambda_1 \geq \lambda_2$. We also use the total structural energy $S = \lambda_1 + \lambda_2$ and the normalized two-directional response

$$\eta = \frac{4\lambda_1\lambda_2}{(\lambda_1 + \lambda_2)^2 + \epsilon}. \quad (2)$$

The main two-qubit QPP-QNN directly encodes $[\lambda_1, \lambda_2]$, while the one-qubit variant uses the compact scalar $t = \log(S + \epsilon) + 4\eta$. These representations compress the local second-order geometry before quantum processing instead of assigning one qubit to every handcrafted feature.

Feature normalization is fitted exclusively on the training split. For a raw feature x_j , the rotation angle is computed as

$$z_j = \text{clip}\left(\frac{x_j - \mu_j}{\sigma_j + \epsilon}, -3, 3\right), \quad \phi_j = \frac{\pi}{3}z_j, \quad (3)$$

and the same training statistics are then applied to the validation and test sets. The QPP-QNNs use two data-reuploading layers. Each layer applies $R_Y(\phi_j)R_Z(\phi_j)$ encoding followed by trainable $R_ZR_YR_Z$ rotations. The two-qubit model additionally applies a CNOT($0 \rightarrow 1$) gate and measures $\langle Z_0 \rangle$, $\langle Z_1 \rangle$, and $\langle Z_0Z_1 \rangle$; the one-qubit model measures $\langle Z \rangle$. A trainable affine readout followed by a sigmoid maps these expectation values to the keypoint probability. Unless explicitly stated otherwise, circuit expectations are evaluated exactly with the differentiable state-vector simulator, without finite-shot sampling.

1.1.3 Compared Methods and Training Settings

The reference methods cover three complementary groups. First, Harris, FAST, and ORB are included as conventional image-level keypoint detectors. Harris is run with block size 2, Sobel aperture 3, $k = 0.04$, and a relative response threshold of 0.01; FAST uses threshold 20 with non-maximum suppression; and each detector retains at most 40 points per image. Second, the classical learning reference is a multilayer perceptron with hidden widths 32 and 16, ReLU activations, and the five-dimensional structure-tensor input $[I_x, I_y, \lambda_1, \lambda_2, R]$. Scalar λ_2 thresholding and logistic regression on $[\log(S + \epsilon), \eta]$ are also retained as compact controls. Third, the quantum comparison includes the proposed one- and two-qubit QPP-QNNs as well as the earlier feature-dimension-matched QNN families used in the ablation study.

The MLP is standardized using training-set statistics and optimized with Adam for at most 300 iterations, using a learning rate of 0.003 and a batch size of 128. The QPP-QNN models are trained on CPU using Adam, a learning rate of 0.01, a batch size of 128, and at most 40 epochs. A class-weighted binary cross-entropy loss compensates for the 1:4 positive-to-negative ratio, and early stopping with patience 8 monitors validation PR-AUC. QPP-QNN training uses random seed 47. For the QPP-QNNs and compact score-based controls, the decision threshold is selected on the validation split by maximizing F1 and is then fixed before evaluating the test split. The archived MLP reference uses the conventional probability threshold of 0.5. Configuration changes made for individual noise, low-sample, ablation, or hardware experiments are stated in their corresponding subsections.

1.1.4 Evaluation Metrics

Because only 20% of the candidate patches are positive, accuracy alone can overstate performance. We therefore use F1 and precision-recall area under the curve (PR-AUC) as the primary measures and report precision and recall to show the false-positive/false-negative trade-off. With true positives TP , false positives FP , and false negatives FN , these metrics are

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad (4)$$

and

$$F_1 = 2 \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (5)$$

PR-AUC is implemented as average precision rather than trapezoidal interpolation. For operating points indexed in descending score order, it is computed as

$$\text{PR-AUC} = \text{AP} = \sum_n (R_n - R_{n-1}) P_n, \quad (6)$$

where P_n and R_n denote precision and recall at the n th threshold. Accuracy and ROC-AUC are retained as secondary diagnostics. Patch classifiers are scored at the candidate level. For the conventional image detectors, a detected point is counted as a true positive when it can be uniquely matched to an unmatched ground-truth keypoint within a three-pixel tolerance; all remaining detections and annotations are counted as false positives and false negatives, respectively.

1.2 Main Results on the Clean Benchmark

We first examine whether the compact phase-coordinate QPP-QNN retains enough local geometric information to distinguish keypoints from non-keypoints under clean imaging conditions. The comparison includes three conventional full-image detectors, a shared-feature MLP, and the proposed one- and two-qubit QPP-QNNs. Table 1 reports the complete clean-test metrics, while Fig. 1 provides a direct visual comparison of their F1 scores.

1.2.1 Overall Clean-Test Comparison

Table 1: Clean synthetic corner/junction results for the classical references and phase-coordinate QPP-QNNs.

Method	Input	Precision	Recall	F1	PR-AUC
Harris	image	0.1652	0.9500	0.2815	0.8953
FAST	image	0.0789	0.7833	0.1433	0.7925
ORB	image	0.0412	1.0000	0.0791	0.7759
MLP	shared 5D features	0.9347	0.9067	0.9205	0.9749
QPP-QNN 1q L2	scalar- $c4$	0.8679	0.9633	0.9131	0.9437
QPP-QNN 2q L2	$[\lambda_1, \lambda_2]$	0.8865	0.9633	0.9233	0.9448

The two groups in Table 1 use related but distinct operating protocols. For Harris, FAST, and ORB, precision, recall, and F1 are obtained by one-to-one spatial matching within three pixels, whereas PR-AUC is computed by ranking their response values at the sampled candidate centers. For the MLP and QPP-QNNs, all metrics are computed directly from candidate classification scores. The full-image detectors should therefore be regarded as operational image references rather than parameter- or input-matched learners. Comparisons within the learned-model group are more direct because these models are evaluated on the same local candidates and image-level split.

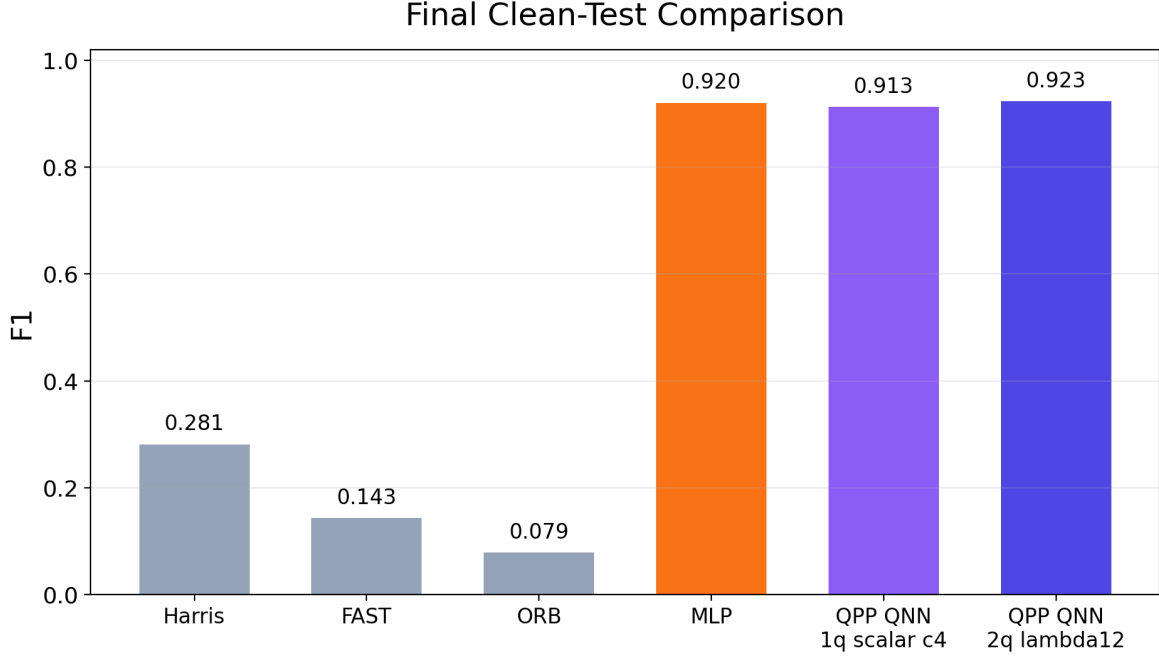


Figure 1: Clean-test F1 comparison among the full-image classical detectors (Harris, FAST, and ORB), the shared-feature MLP, and the one- and two-qubit phase-coordinate QPP-QNNs. The classical detectors are evaluated by spatial point matching, whereas the learned models classify the same sampled local candidates.

Figure 1 shows a clear separation between the conventional detectors and the learned local classifiers on this sparse geometric benchmark. It also shows that both few-qubit models operate in the same F1 regime as the MLP despite using only one or two geometric coordinates as quantum inputs. The following comparisons examine these two observations in more detail.

1.2.2 Comparison with Classical References

Among the full-image detectors, Harris gives the strongest F1 score of 0.2815, followed by FAST at 0.1433 and ORB at 0.0791. Their recalls remain relatively high—0.9500, 0.7833, and 1.0000, respectively—but their precisions fall to 0.1652, 0.0789, and 0.0412. Thus, these methods locate most annotated junctions but also produce many unmatched responses on the sparse line drawings. This behavior is consistent with their local response rules: strong gradient changes can generate detections even when the response does not correspond to the labeled candidate geometry.

The MLP is a substantially stronger classical learning reference. Using the shared five-dimensional structure-tensor features, it reaches 0.9347 precision, 0.9067 recall, 0.9205 F1, and 0.9749 PR-AUC. Its high PR-AUC indicates that its scores rank positive candidates consistently over a wide range of decision thresholds. The large gap between the MLP and the full-image detectors should not, however, be attributed solely to the model family: the MLP benefits from candidate-level supervision, while Harris, FAST, and ORB operate directly on the image without access to the learned patch labels. The MLP therefore serves as the principal classical comparator for judging the learned QPP models.

1.2.3 Comparison between One- and Two-Qubit QPP Models

The one-qubit QPP-QNN compresses the local spectrum into $t = \log(S + \epsilon) + 4\eta$ and achieves 0.8679 precision, 0.9633 recall, 0.9131 F1, and 0.9437 PR-AUC with two re-uploading layers.

This result shows that a single-qubit circuit driven by one geometry-informed scalar already captures a useful keypoint decision boundary. The two-qubit model instead retains the ordered eigenvalues $[\lambda_1, \lambda_2]$ as separate phase coordinates. It preserves the same 0.9633 recall while increasing precision to 0.8865, which raises F1 to 0.9233. Relative to the one-qubit model, this is an absolute F1 improvement of 0.0102, whereas PR-AUC changes only slightly from 0.9437 to 0.9448. The improvement is therefore mainly associated with a better fixed operating point rather than a large change in global score ranking.

Compared with the MLP, the archived two-qubit result has a marginally higher F1 (0.9233 versus 0.9205) and higher recall (0.9633 versus 0.9067), but lower precision (0.8865 versus 0.9347) and lower PR-AUC (0.9448 versus 0.9749). Moreover, as described in Section 1.1, the archived MLP uses a threshold of 0.5, whereas the QPP-QNN threshold is selected on the validation split. We therefore interpret the result as evidence that a shallow two-qubit phase-coordinate model is competitive with the classical learner at the selected operating point, rather than as evidence of an overall quantum advantage. Its main benefit in this experiment is resource efficiency: the model retains strong clean-test detection performance using two qubits, two re-uploading layers, and a compact spectral input.

1.3 Separating L-Corners, T-Junctions, and X-Junctions

The binary benchmark in Section 1.2 treats L-corners, T-junctions, and X-junctions as a single positive keypoint class. We next ask whether the same local structure-tensor representation retains enough geometric information to distinguish these three structures from one another. This experiment is therefore a three-class classification task conditioned on patches that contain an annotated keypoint; it is not a four-class detector with an additional background category. Besides testing finer geometric semantics, this setting allows a compact classical-quantum comparison under a common input representation and a group-disjoint evaluation protocol.

1.3.1 Multiclass Protocol and Evaluation Metrics

The dataset contains 120 independently rendered source-image groups for each of the L-, T-, and X-junction classes. Four positive patches are sampled from each group, yielding $120 \times 4 \times 3 = 1,440$ labeled patches in total. A stratified 60/20/20 split is performed at the source-group level, resulting in 864 training, 288 validation, and 288 test patches. Group-level separation is essential because patches derived from the same rendered source can share nearly identical geometry and appearance; allowing them to cross subsets would therefore lead to an overly optimistic estimate of generalization.

All learners receive the same eight-dimensional structure-tensor interface,

$$\mathbf{x} = [I_x, I_y, I_x^2, I_y^2, I_x I_y, \lambda_1, \lambda_2, R], \quad (7)$$

and feature normalization is fitted only on the training split. Results are aggregated over five training seeds. For models that are retrained for every seed, we report the sample mean and standard deviation; deterministic reference values are reported once when no run-to-run variation is available.

Because the task contains three target classes, we use macro averages. For a class $c \in \{L, T, X\}$, that class is treated as positive and the other two as negative, giving

$$P_c = \frac{TP_c}{TP_c + FP_c}, \quad R_c = \frac{TP_c}{TP_c + FN_c}, \quad F_{1,c} = 2 \frac{P_c R_c}{P_c + R_c}. \quad (8)$$

Macro-F1 assigns equal importance to the three geometric categories:

$$\text{Macro-F1} = \frac{1}{3} \sum_{c \in \{L, T, X\}} F_{1,c}. \quad (9)$$

Likewise, a one-vs-rest precision–recall curve is constructed for each class, and the three classwise areas are averaged:

$$\text{Macro-PR-AUC} = \frac{1}{3} \sum_{c \in \{L, T, X\}} \text{PR-AUC}_c. \quad (10)$$

These macro averages prevent performance on one comparatively easy class from dominating the reported score and directly assess whether a model recognizes all three junction types consistently.

1.3.2 Compared Classical and Quantum Models

Four models are compared under the shared feature and split protocol. The multinomial logistic model is the linear reference, with class probabilities

$$\mathbf{p}(y \mid \mathbf{x}) = \text{softmax}(\mathbf{W}\mathbf{x} + \mathbf{b}). \quad (11)$$

For eight inputs and three output classes, it contains $8 \times 3 + 3 = 27$ trainable parameters and measures the linear separability of the junction types. The RBF SVM supplies a stronger nonlinear classical reference through the kernel

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|_2^2). \quad (12)$$

Its complexity depends on the selected support vectors, so it is listed as nonparametric rather than assigned a fixed parameter count.

TinyMLP h5 uses a single hidden layer of width five between the eight inputs and the three-class output. Its parameter count is

$$(8 \times 5 + 5) + (5 \times 3 + 3) = 63, \quad (13)$$

making it a compact nonlinear neural baseline. The quantum comparator is a four-qubit, two-layer grouped data-reuploading QNN with 67 trainable parameters. Across its two layers, the eight input coordinates are distributed over four qubits, followed by trainable local rotations, ring entanglers, Pauli readout, and an affine three-class head. The 63-parameter TinyMLP and 67-parameter QNN therefore provide an approximately parameter-matched classical–quantum comparison, although their computational primitives remain fundamentally different.

1.3.3 Results and Discussion

Table 2: Three-class L/T/X classification under a source-image-disjoint split. Neural and kernel results are aggregated over five seeds.

Model	Parameters	Macro-F1	Macro PR-AUC
Multinomial logistic	27	0.7719	0.7942
RBF SVM	nonparametric	0.7700 ± 0.0018	0.8656 ± 0.0004
TinyMLP h5	63	0.8077 ± 0.0238	0.8206 ± 0.0140
Grouped QNN 4q L2	67	0.7636 ± 0.0232	0.7934 ± 0.0122

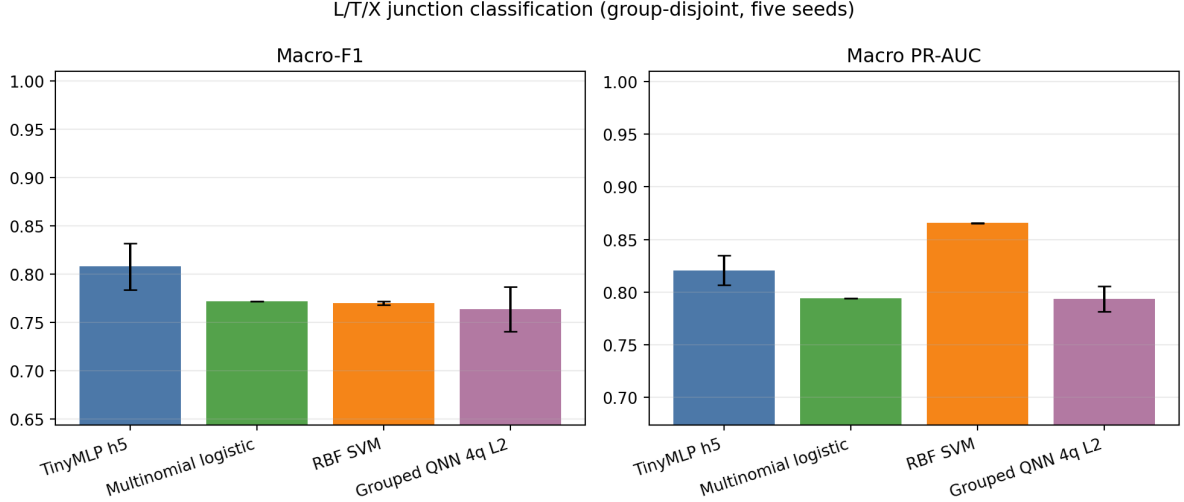


Figure 2: Five-seed L/T/X classification under a group-disjoint split. Bars show mean Macro-F1 and Macro PR-AUC, and error bars show standard deviations where repeated-run estimates are available. TinyMLP achieves the highest Macro-F1, whereas the RBF SVM achieves the highest Macro PR-AUC.

As shown in Table 2 and Fig. 2, TinyMLP gives the best final classification balance, reaching a Macro-F1 of 0.8077 ± 0.0238 . Its advantage over multinomial logistic regression indicates that the junction types are not fully separated by linear decision boundaries in the shared eight-dimensional feature space. The RBF SVM attains a lower Macro-F1 of 0.7700 ± 0.0018 but the highest Macro PR-AUC of 0.8656 ± 0.0004 . It therefore provides the most consistent one-vs-rest score ranking across the three classes, with very little variation across runs, even though its final multiclass decisions do not maximize Macro-F1.

The grouped four-qubit QNN reaches 0.7636 ± 0.0232 Macro-F1 and 0.7934 ± 0.0122 Macro PR-AUC. These values demonstrate nontrivial learning of L/T/X geometry and are close to the 27-parameter multinomial logistic reference, but they remain below the best classical results. In particular, the approximately parameter-matched TinyMLP exceeds the QNN by 0.0441 in Macro-F1, while the RBF SVM exceeds it by 0.0722 in Macro PR-AUC. The present experiment therefore supports task extensibility rather than quantum superiority: the local PSD representation and grouped QNN can be extended from binary keypoint presence to finer junction semantics, but the classical TinyMLP and RBF SVM remain stronger under the current protocol. This outcome also motivates the controlled quantum-family study in the next subsection, where the input, qubit count, training budget, and threshold protocol are held fixed more tightly.

1.4 Few-Qubit Quantum-Family Comparison

Having established the feasibility of spectrum-based quantum classification, we next ask a more focused question: which quantum encoding and circuit family offers the most favorable performance under the same small-qubit budget? This experiment is an intra-quantum comparison rather than a claim of quantum advantage. Its purpose is to separate the effects of state preparation, variational-circuit structure, and parameter count while keeping the classical input and evaluation procedure fixed.

1.4.1 Controlled Comparison Protocol

All models receive the same ordered tensor-spectrum feature $\mathbf{x} = [\lambda_1, \lambda_2]$, use the same image-level train/validation/test split and feature normalization, and are evaluated on the same clean

test set. The trainable circuits use two qubits and two variational layers. We also hold the optimizer, training budget, and validation-based threshold-selection rule fixed. F1 and PR-AUC are computed using the definitions in Section 1.1.4; the reported values are the mean and standard deviation over multiple random seeds. Consequently, the comparison reflects differences among the quantum model families rather than changes in data partitioning or evaluation protocol.

The two fidelity-kernel baselines shown in Fig. 3 use the same inputs and split, but they are not variational QNNs. They construct a fixed quantum similarity matrix and pass it to a classical kernel learner. They are therefore included as references in the figure, but excluded from the trainable-parameter comparison in Table 3.

1.4.2 Quantum Model Families

The comparison covers complementary ways of embedding and processing the two spectral coordinates:

- **Schmidt-QPP QNN.** The normalized spectrum $\mu_i = \lambda_i/(\lambda_1 + \lambda_2)$ is encoded as the amplitudes of a correlated two-qubit state,

$$|\psi_S(\mathbf{x})\rangle = \sqrt{\mu_1}|00\rangle + \sqrt{\mu_2}|11\rangle. \quad (14)$$

A shallow parameterized circuit and Pauli-expectation readout then learn the decision function. This construction places the relative local spectrum directly in a correlated subspace.

- **Data re-uploading QNN.** The normalized coordinates are repeatedly injected as single-qubit rotation angles at every layer, with trainable local rotations and a shallow entangling operation between successive uploads. Repeated access to the data can increase the non-linearity of a shallow circuit without increasing the qubit count.
- **HEA-VQC with angle encoding.** A conventional hardware-efficient ansatz combines angle encoding, trainable local rotations, and shallow two-qubit entanglers. It provides a generic variational baseline that is not tailored specifically to the tensor spectrum.
- **Separable VQC with angle encoding.** This control retains local angle encoding and trainable one-qubit transformations but removes the two-qubit entangling operation. Its comparison with the entangling circuits tests whether independent processing of λ_1 and λ_2 is sufficient under the same shallow-depth regime.
- **IQP-VQC.** This model uses an IQP-style phase feature map followed by a trainable variational readout. It tests whether a phase-interaction encoding is well matched to the ordered spectral input.

For completeness, the two fixed quantum-kernel references use either an IQP-based feature state or the Schmidt state in Eq. (14). Their kernel entries are

$$K(\mathbf{x}_i, \mathbf{x}_j) = |\langle \psi(\mathbf{x}_i) | \psi(\mathbf{x}_j) \rangle|^2, \quad (15)$$

so their learnable decision boundary is produced by the classical kernel learner rather than by optimizing quantum-circuit parameters.

1.4.3 Results and Resource–Performance Trade-offs

Table 3: Controlled comparison of trainable two-qubit model families on the clean test set. Results are reported as mean $\pm$ standard deviation over multiple random seeds. Bold entries denote the best value in each metric.

Model	Trainable parameters	F1	PR-AUC
Schmidt-QPP QNN, $L = 2$	25	0.9279 ± 0.0045	0.9539 ± 0.0065
Data re-uploading QNN, $L = 2$	16	0.9181 ± 0.0099	0.9470 ± 0.0072
HEA-VQC angle, $L = 2$	16	0.9179 ± 0.0088	0.9650 ± 0.0025
Separable VQC angle, $L = 2$	16	0.8961 ± 0.0080	0.9540 ± 0.0061
IQP-VQC, $L = 2$	16	0.8574 ± 0.0404	0.9092 ± 0.0338

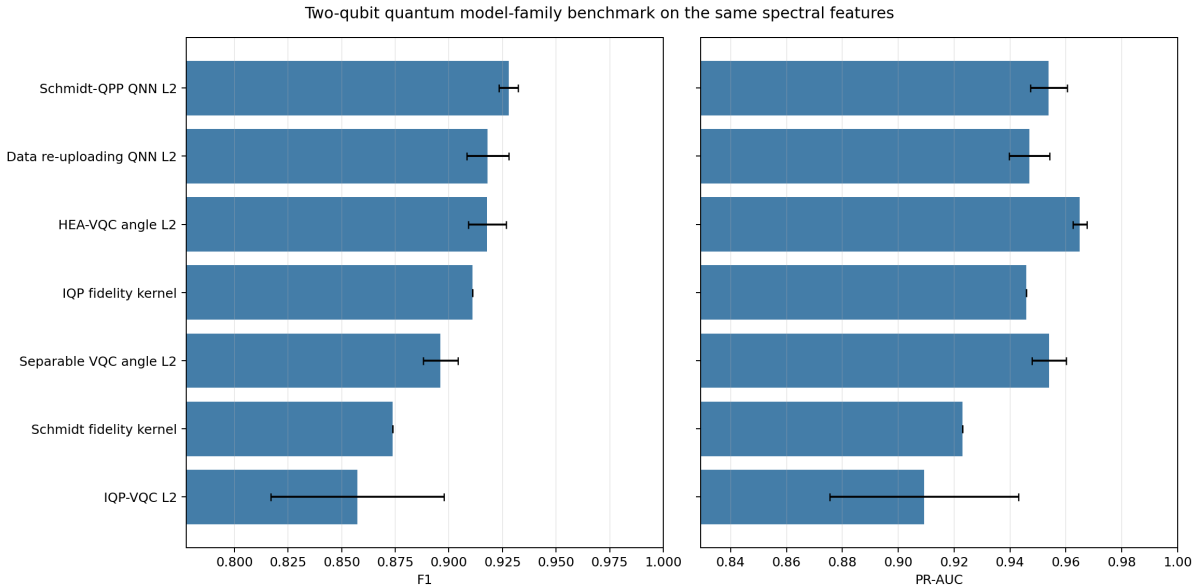


Figure 3: F1 (left) and PR-AUC (right) for the controlled two-qubit family comparison. The five trainable QNN/VQC models correspond to Table 3; the IQP and Schmidt fidelity kernels are fixed-embedding quantum-kernel references and therefore have no comparable variational parameter count. Error bars indicate variability across repeated estimates where available.

The Schmidt-QPP QNN obtains the highest F1, 0.9279 ± 0.0045 , indicating the best balance between precision and recall at the validation-selected operating threshold. The data re-uploading QNN is a close second at 0.9181 ± 0.0099 while using 16 rather than 25 trainable parameters—a 36% reduction. It therefore provides an attractive resource–performance compromise when parameter count and hardware-oriented circuit simplicity are important.

The ranking changes when threshold-independent quality is considered. The HEA-VQC achieves the highest PR-AUC, 0.9650 ± 0.0025 , although its F1 is below that of Schmidt-QPP. Thus, Schmidt-QPP provides the strongest result at the selected decision threshold, whereas HEA-VQC gives the best overall precision–recall ranking across thresholds. Reporting both metrics prevents either behavior from being hidden by a single operating point.

The separable VQC retains a competitive PR-AUC of 0.9540 ± 0.0061 but its F1 drops to 0.8961 ± 0.0080 . This result suggests that independent one-qubit processing can rank many samples correctly, yet is less effective at forming a balanced decision boundary at the selected threshold. Because the ansatzes differ in more than entanglement alone, this gap should be interpreted as evidence for the benefit of joint spectral modeling in the tested setting, not as

an isolated causal demonstration of quantum entanglement. IQP-VQC records the lowest mean scores and the largest run-to-run variation (0.8574 ± 0.0404 F1), showing that the tested IQP-style trainable encoding is both less accurate and less stable for these two ordered eigenvalue features.

Figure 3 further shows that the two fixed fidelity-kernel references do not dominate the best trainable circuits. A fixed quantum similarity geometry alone is therefore insufficient to explain the leading results; task-adaptive variational training is important in this benchmark. Overall, no single family wins every criterion: Schmidt-QPP is the preferred accuracy-oriented model, HEA-VQC provides the strongest PR-AUC, and data re-uploading offers the clearest reduction in trainable parameters with only a modest F1 loss. These conclusions establish a resource-aware basis for the noise-robustness experiments in the following section.

1.5 Noise Robustness

We next evaluate whether the clean-data decision rules remain usable after the image distribution changes. These experiments concern corruption of the *input images*, rather than gate or measurement noise on quantum hardware. Unless explicitly stated otherwise, every learner is trained on clean images, its decision threshold is selected on the clean validation split, and that threshold is then frozen for all corrupted test sets. Noise is applied to the held-out test images before the original candidate centers are used to extract new 9×9 patches and recompute the structure-tensor features. The protocol therefore measures robustness without test-time threshold recalibration.

The evidence below comes from four deliberately separated study blocks. The first reproduces the original phase-coordinate QPP-QNN corruption test. The second is an additional matched-feature study of the earlier 5D/8D QNN families against corresponding MLPs. The third is a newly reconstructed, scale-free Schmidt diagnostic designed specifically to isolate the effect of normalizing away tensor energy. The final block reports noise-aware adaptation for the original phase-coordinate branch and the archived Schmidt branch. Results are compared only within a block unless the models, inputs, and training protocol are explicitly matched; absolute numbers from different blocks should not be interpreted as repeated runs of one model.

In addition to absolute F1, we use the clean-relative F1 retention

$$\rho_{\text{F1}}(\nu) = \frac{\text{F1}(\nu)}{\text{F1}(0)}, \quad (16)$$

where ν denotes the corruption intensity. Absolute F1 determines which model is preferable at a given operating condition, whereas ρ_{F1} separates degradation rate from clean-set accuracy. PR-AUC is reported alongside both measures because a model can lose its fixed operating point while still retaining useful score ordering.

1.5.1 Gaussian and Blur Corruption

Figure 4 first reports the original compact phase-coordinate study under two levels of additive Gaussian noise, Gaussian blur, and salt-and-pepper corruption. Pixel-level Gaussian noise is generated as $I' = \text{clip}(I + \epsilon, 0, 1)$ with $\epsilon \sim \mathcal{N}(0, \sigma^2)$. The two-qubit QPP-QNN is particularly stable under mild additive corruption: its F1 changes from 0.9204 on clean images to 0.9080 at $\sigma = 0.04$, corresponding to 98.7% retention. At $\sigma = 0.08$ it still reaches 0.8266 F1 and 0.9254 PR-AUC, retaining 89.8% of its clean F1. By contrast, the one-qubit scalar model falls from 0.9131 to 0.3989 F1 at the same noise level. The large separation at $\sigma = 0.08$ shows that preserving $[\lambda_1, \lambda_2]$ as two coordinates is much more reliable than compressing the spectrum into the single scalar $\log(S + \epsilon) + 4\eta$ under this perturbation.

The advantage is corruption dependent rather than universal. Under blur radius 0.9, the two-qubit model obtains 0.6797 F1: its precision remains high at 0.9691, but recall decreases to

0.5233 because smoothing attenuates the local gradients used by the fixed clean threshold. The one-qubit model and the logistic $[\log(S + \epsilon), \eta]$ control reach 0.8100 and 0.9000 F1, respectively. Thus, the two-coordinate circuit provides the strongest QPP response to Gaussian perturbation, but blur remains a distinct failure mode that cannot be summarized by a general claim of corruption invariance.

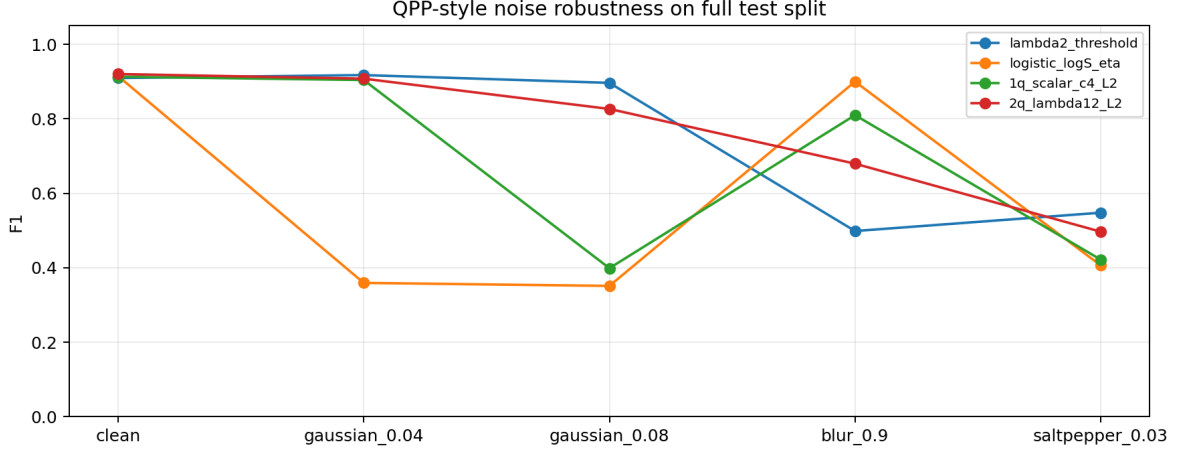


Figure 4: Fixed-threshold F1 of the phase-coordinate QPP-QNNs and compact classical controls under clean, Gaussian, blur, and salt-and-pepper test conditions. All thresholds are selected before corrupted test evaluation. The two-qubit $[\lambda_1, \lambda_2]$ model remains substantially more stable than the one-qubit scalar encoding at Gaussian $\sigma = 0.08$.

1.5.2 Extended Matched-Feature Salt-and-Pepper Sweep

Salt-and-pepper noise is more challenging because isolated intensity impulses create strong local gradients and can turn background patches into apparent keypoints. We therefore add a separate stress test based on the earlier feature-dimension QNN families. This is an extension of the original phase-coordinate experiment above, not a new noise evaluation of the same one- or two-qubit circuit. To compare classical and quantum learners within this extension under matched inputs, we use a five-dimensional interface $[I_x, I_y, \lambda_1, \lambda_2, R]$ and an eight-dimensional interface $[I_x, I_y, I_x^2, I_y^2, I_x I_y, \lambda_1, \lambda_2, R]$. Within each feature dimension the QNN and MLP use the same clean train/validation/test patches, and the noise probability is swept from $p = 0$ to $p = 0.15$. Their clean scores are therefore valid baselines for the retention calculation in this block, but are not substitutes for the clean phase-coordinate results in Section 1.5.1.

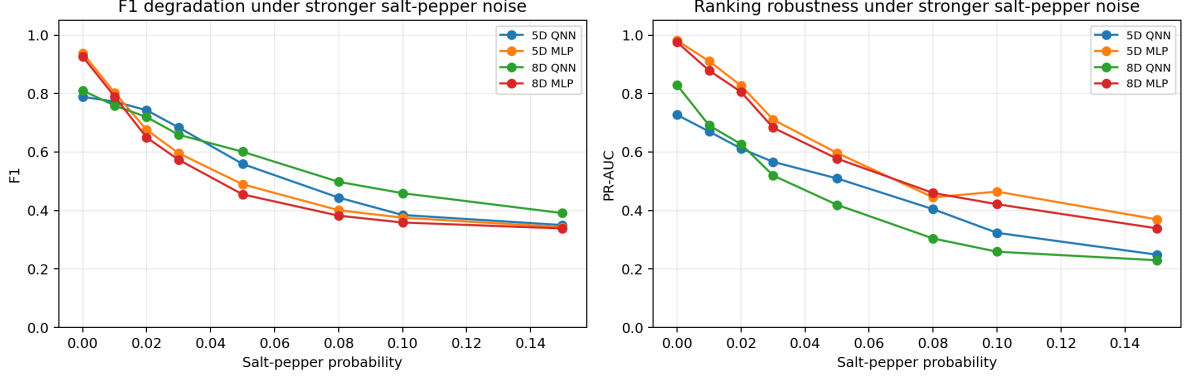


Figure 5: Additional matched-feature comparison of the earlier five- and eight-dimensional QNN/MLP families under increasing salt-and-pepper probability. The left panel reports F1 at the frozen clean threshold and the right panel reports PR-AUC. The QNN curves decay more slowly in fixed-threshold F1, whereas the MLPs retain stronger global score ranking over much of the sweep.

Figure 5 reveals a crossover between clean accuracy and corrupted operating-point stability. The MLPs are stronger on clean images, reaching 0.9430 and 0.9338 F1 for the 5D and 8D inputs, respectively, compared with 0.7890 and 0.8105 for the corresponding QNNs. At $p = 0.01$ the MLPs retain a small absolute lead. From $p = 0.02$ onward, however, both QNNs obtain higher fixed-threshold F1 than their dimension-matched MLPs. The common $p = 0.03$ operating point is summarized in Table 4.

Table 4: Matched QNN/MLP performance under salt-and-pepper probability $p = 0.03$. Retention is defined by Eq. (16); precision and PR-AUC are measured on the corrupted test set.

Model	Clean F1	Noisy F1	F1 retention	Precision	PR-AUC
5D QNN	0.7890	0.6842	0.867	0.5482	0.5665
5D MLP	0.9430	0.5774	0.612	0.4123	0.7048
8D QNN	0.8105	0.6593	0.813	0.5224	0.5198
8D MLP	0.9338	0.5611	0.601	0.3921	0.6775

At $p = 0.03$, the 5D QNN exceeds the 5D MLP by 0.1068 F1 and retains 86.7% of its clean score, versus 61.2% for the MLP. The 8D QNN similarly exceeds its MLP by 0.0982 F1 and retains 81.3%, versus 60.1%. The MLPs maintain recall above 0.96 but their precision falls to 0.4123 and 0.3921, indicating that impulse-induced false positives overwhelm the clean-calibrated decision threshold. The QNNs trade a modest amount of recall for precision above 0.52 and consequently obtain the better deployed F1 without recalibration.

Over the complete $p = 0$ to $p = 0.15$ sweep, F1 decreases by 0.4387 for the 5D QNN and 0.4193 for the 8D QNN, compared with drops of 0.6020 and 0.5930 for the corresponding MLPs. Nevertheless, the right panel of Fig. 5 shows that MLP PR-AUC remains higher over most noise levels. The supported advantage is therefore precise: the QNNs provide a more stable *fixed clean-calibrated operating point* under impulsive corruption, rather than a uniformly superior ranking at every possible threshold.

1.5.3 Reconstructed Scale-Free Schmidt Diagnostic

The preceding results use feature sets that retain gradient magnitude and tensor-energy information. We next isolate what happens when the quantum state contains only the normalized Schmidt spectrum. Because the training script and checkpoint associated with the original

archived Schmidt noise table are not available, the following experiment is explicitly treated as a new reconstruction rather than a replication of that table. The matched diagnostic compares a 25-parameter scale-free Schmidt reconstruction with two 21-parameter TinyMLPs. The first MLP receives the raw ordered eigenvalues $[\lambda_1, \lambda_2]$, while the second receives exactly the normalized spectrum $[\mu_1, \mu_2]$ encoded by the Schmidt state, where $\mu_i = \lambda_i/(\lambda_1 + \lambda_2)$. Results are averaged over five training seeds and, for nonzero corruption, three independently generated noise instances per training seed.

Table 5: F1 in the independently reconstructed, representation-matched robustness diagnostic. The raw-spectrum MLP retains tensor energy, whereas the scale-free Schmidt model and the μ_{12} MLP are invariant to a common rescaling of both eigenvalues.

Test condition	Scale-free Schmidt	MLP λ_{12}	MLP μ_{12}
Clean	0.8738	0.9259	0.8738
Gaussian $\sigma = 0.01$	0.3586	0.9250	0.3586
Gaussian $\sigma = 0.04$	0.3542	0.9147	0.3542
Salt-and-pepper $p = 0.03$	0.3730	0.4843	0.3730

The scale-free Schmidt reconstruction and the representation-matched μ_{12} MLP produce identical F1, precision, and recall throughout both noise sweeps, with only numerical round-off differences in PR-AUC. In contrast, the raw-spectrum MLP is far more stable under Gaussian noise. The reason is visible in the feature statistics. For clean negative patches the median tensor energy $S = \lambda_1 + \lambda_2$ and median μ_2 are both zero. At $\sigma = 0.01$, the negative-patch median energy is still only 0.0141, compared with 5.1257 for positive patches, but its median μ_2 rises to 0.4202, slightly above the positive median of 0.3746. Normalization therefore makes weak, nearly isotropic noise resemble a genuine corner while discarding the small energy that would identify it as background.

This diagnostic does not imply that every quantum encoding is noise sensitive; it identifies a boundary of this reconstructed scale-free representation. In particular, its values must not be substituted for either the clean Schmidt family comparison in Table 3 or the archived Schmidt noise-aware summary below, because those results use different circuit and training implementations. The diagnostic instead shows why the positive 5D/8D and phase-coordinate results should be attributed to the joint choice of retained geometric information and classifier, not to quantumness alone. It further motivates an energy-aware Schmidt interface, such as $[\log(S + \epsilon), \mu_2]$, an explicit trace floor, or a classical energy gate alongside the quantum readout.

1.5.4 Noise-Aware Training

Finally, we return to the original study and ask whether its difficult salt-and-pepper operating points can be recovered by exposing the QPP-QNNs to representative corruptions during optimization. For the reproducible phase-coordinate run, clean-only training uses 4,500 patches. The noise-aware run augments them with Gaussian $\sigma \in \{0.04, 0.08\}$, blur radius 0.9, and salt-and-pepper $p = 0.03$ versions, giving 22,500 training instances. The corrupted images are passed through the complete feature-extraction pipeline rather than perturbing the already computed feature vectors. The initial study also archives summary metrics for a separate Schmidt-QPP branch. Those Schmidt values are reported as their own row pair and are not combined with the reconstructed scale-free diagnostic in Section 1.5.3.

Table 6: Within-branch recovery on salt-and-pepper $p = 0.03$ before and after noise-aware training. The phase-coordinate row is reproduced from the available run outputs; the Schmidt row records the archived summary from the initial study.

Model branch	F1 clean	F1 adapted	PR-AUC clean	PR-AUC adapted
Phase-coordinate 2q	0.5448	0.7307	0.4857	0.7712
Archived Schmidt-QPP	0.5064	0.7128	0.5530	0.8146

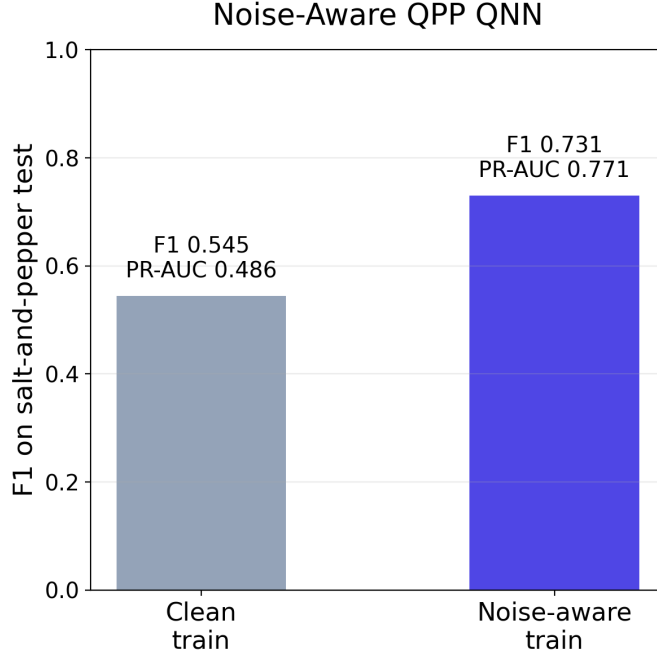


Figure 6: Recovery of the phase-coordinate two-qubit QPP-QNN under salt-and-pepper $p = 0.03$. This figure visualizes the first row of Table 6; the archived Schmidt branch is kept numerically separate in the second row.

For the phase-coordinate branch, noise-aware training increases F1 by 0.1859, from 0.5448 to 0.7307, which is a 34.1% relative improvement. PR-AUC rises by 0.2855, from 0.4857 to 0.7712. The available detailed output shows that the dominant operating-point change is the precision increase from 0.3768 to 0.6407: although recall decreases from 0.9833 to 0.8500, the model suppresses enough impulse-induced false positives to achieve a substantially better precision-recall balance.

The archived Schmidt branch exhibits the same qualitative adaptation pattern: F1 increases by 0.2064, from 0.5064 to 0.7128, and PR-AUC increases by 0.2616, from 0.5530 to 0.8146. These summary values support the original observation that impulse noise requires augmentation, but they are not evidence that the archived Schmidt implementation and the new scale-free reconstruction behave identically. Moreover, both noise-aware branches use augmented training data and neither has a matched noise-aware MLP in this study. The result therefore demonstrates recoverability of each QPP-QNN branch rather than a matched classical advantage.

Taken together, the separated experiments establish a conditional robustness result. In the original phase-coordinate study, the two-qubit spectral representation is substantially more stable than its one-qubit compression under Gaussian noise. In the additional matched-feature study, the 5D/8D QNNs maintain a more stable clean-calibrated operating point than the matched MLPs under moderate salt-and-pepper corruption. The newly reconstructed scale-free

diagnostic then shows that this behavior is representation dependent: removing the tensor-energy cue can eliminate robustness even when the downstream circuit is trainable. Finally, the phase-coordinate and archived Schmidt branches both improve substantially with noise-aware training. Robust quantum-edge inference should therefore be understood as the joint design of geometric representation, circuit architecture, and corruption-aware training, rather than as an automatic consequence of using a quantum classifier.

1.6 Ablation Studies

1.6.1 Circuit Depth

1.6.2 Entangling Patterns

1.6.3 Observable Readout

1.7 Cross-Domain Transfer

1.7.1 Three-Dimensional Point Covariance

1.7.2 Multichannel PSD Classification

1.8 Quantum Hardware Execution

1.8.1 Backend and Circuit Configuration

1.8.2 Readout Correction and Hardware Results